

ResNet and DenseNet

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1 Background

1.1 Limitation of feed forward network

More layers, more nodes, and more parameters can improve representation and accuracy. However, a degradation problem may occur in deep networks because of gradient vanishing, explosion, and dead neurons [1].

1.1.1 Gradient vanish, explosion and dead neuron

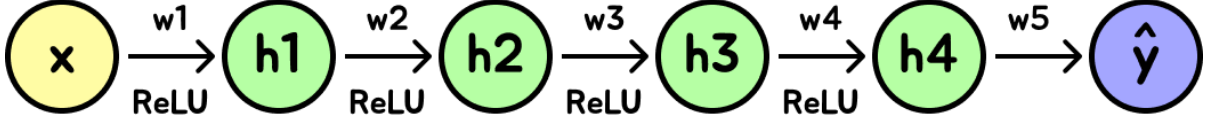


Figure 1: Example of simple network

In a simple network in Figure 1, the model returns y , defined by Equation 1, where the activation function, f , is ReLU.

$$\hat{y} = w_5 f(w_4 f(w_3 f(w_2 f(w_1 x)))) \quad (1)$$

From the backpropagation perspectives of w_1 , the gradient with MSE loss (mean square error) of ground truth, y , is defined by Equation 2, where $h_0 = x$.

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h_4} \cdot \frac{\partial h_4}{\partial h_3} \cdot \frac{\partial h_3}{\partial h_2} \cdot \frac{\partial h_2}{\partial h_1} \cdot \frac{\partial h_1}{\partial w_1} = 2(y - \hat{y})x \prod_{i=1}^4 (w_i f'(w_i h_{i-1})) \quad (2)$$

The gradient vanish means the gradient signal can't reach the w_1 , resulting in shallow networks being learned. The gradient explosion causes the gradient magnitude to grow exponentially as it propagates backward, leading to unstable updates and divergence during training Equation 3.

$$\frac{\partial L}{\partial w_1} \approx 0, \quad \left| \prod_{i=1}^4 (w_i f'(w_i h_{i-1})) \right| \ll 1 \quad \left| \frac{\partial L}{\partial w_1} \right| \gg 1, \quad \left| \prod_{i=1}^4 (w_i f'(w_i h_{i-1})) \right| \gg 1 \quad (3)$$

Especially, dead neuron means a neuron returning zero output for different inputs in ReLU activation, reducing representation and zero gradient.

$$h_i = w_i h_{i-1} \leq 0, \quad f(h_i) = f(h_{i+1}) = f(h_{i+2}) = 0, \quad \frac{\partial L}{\partial w_1} = 0 \quad (4)$$

1.1.2 Forward signal degradation

From the perspective of the forward process, forward signal degradation refers to the model producing similar outputs for different inputs due to multiple linear transformations across deep layers in Equation 5. As a result, the network loses the ability to distinguish between inputs, hindering effective learning.

$$\hat{y}_{x_1} \approx \hat{y}_{x_2} \quad \frac{\partial L_{x_1}}{\partial w_1} \approx \frac{\partial L_{x_2}}{\partial w_1} \quad (5)$$

1.1.3 Hardness of identity mapping

Identity mapping means directly passes from the input to the output in Equation 6. It is necessary in deep networks to ensure that some layers can perform "no operation" when deeper layers are added, allowing the deep network to at least preserve the performance of a shallower network. However, due to nonlinear activations, constructing identity mapping is difficult.

$$f(w_i x) = x \quad (6)$$

1.2 Residual and its utility

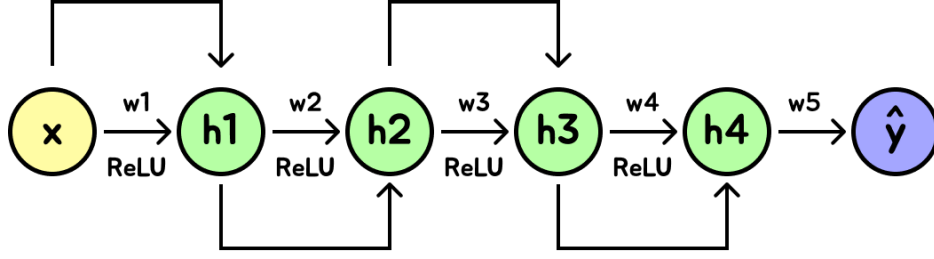


Figure 2: Example of residual network

The concept of residual connection is a composite of activation and its own input defined by Equation 7 in Figure 2. From the backpropagation perspective of w_1 , it can be defined as Equation 7.

$$h_i = f(w_i h_{i-1}) + h_{i-1}, i = 1, 2, 3, 4 \quad y = w_5 h_4, \quad \frac{\partial L}{\partial w_1} = 2(y - \hat{y})w_5 x f'(w_1 x) \prod_{i=1}^4 (w_i f'(w_i h_{i-1}) + 1) \quad (7)$$

Thanks to the residual connections, the value of layer connections parts can be approximately as 1 in the zero mean weight initialization method, preventing the gradient from vanishing and exploding in Equation 8. Furthermore, the model returns different outputs for different inputs, solving the signal degradation and releasing the identity mapping.

$$\left| \prod_{i=1}^4 (w_i f'(w_i h_{i-1}) + 1) \right| \approx 1 \quad \hat{y}_{x_1} \neq \hat{y}_{x_2}, (x_1 \neq x_2) \quad (8)$$

Sometimes, when the input and output sizes differ, the projection matrix, W_s , is applied to the input to match the output dimension. the projection matrix is 1×1 convolution in CNN as Equation 9.

$$h_i = f(W_i h_{i-1}) + W_s h_{i-1}, \quad (h_{i-1} \in \mathbb{R}^{H \times W \times C_{in}}, h_i \in \mathbb{R}^{H \times W \times C_{out}}, W_s \in \mathbb{R}^{1 \times 1 \times C_{in} \times C_{out}}) \quad (9)$$

1.3 Skip connection and its utility

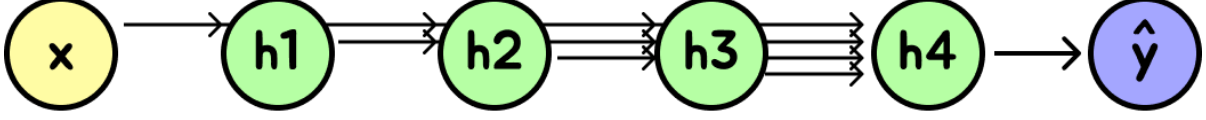


Figure 3: Example of skip connection

The skip connection, referred to as dense connectivity, constructs the output of each layer from the feature maps of all preceding layers, as shown in Equation 10.

$$h_i = f\left(\sum_{k=1}^i w_{i,k} h_{k-1}\right) \quad h_i = f(W_i z_i), z_i = [h_{i-1}, h_{i-2}, \dots, h_0] \quad (10)$$

From the gradient perspectives, the gradient of w_1 is defined by the sum of the multiple path in Equation 11, where $w_{l,j}$ means weight of l layer for j node. Since the gradient is defined the sum of each paths, not the multiple, dense structure may release the gradient vanish, explosion and forward signal degradation problems.

$$\begin{aligned} \frac{\partial L}{\partial w_1} &= \frac{\partial h_1}{\partial w_1} \left(\text{Path}_{h_1 \rightarrow h_2 \rightarrow h_3 \rightarrow h_4 \rightarrow \hat{y}} + \text{Path}_{h_1 \rightarrow h_2 \rightarrow h_4 \rightarrow \hat{y}} + \text{Path}_{h_1 \rightarrow h_3 \rightarrow h_4 \rightarrow \hat{y}} + \text{Path}_{h_1 \rightarrow h_4 \rightarrow \hat{y}} \right) \\ &= \frac{\partial h_1}{\partial w_1} \left(\frac{\partial h_2}{\partial h_1} \frac{\partial h_3}{\partial h_2} \frac{\partial h_4}{\partial h_3} \frac{\partial \hat{y}}{\partial h_4} \frac{\partial L}{\partial \hat{y}} + \frac{\partial h_2}{\partial h_1} \frac{\partial h_4}{\partial h_2} \frac{\partial \hat{y}}{\partial h_4} \frac{\partial L}{\partial \hat{y}} + \frac{\partial h_3}{\partial h_1} \frac{\partial h_4}{\partial h_3} \frac{\partial \hat{y}}{\partial h_4} \frac{\partial L}{\partial \hat{y}} + \frac{\partial h_4}{\partial h_1} \frac{\partial \hat{y}}{\partial h_4} \frac{\partial L}{\partial \hat{y}} \right) \\ &= \frac{\partial h_1}{\partial w_1} \sum_{\ell=2}^4 \sum_{\substack{p:\text{paths} \\ \ell \in p}} \left(\prod_{j \in p \text{ after } \ell} \frac{\partial h_j}{\partial h_{\text{prev}(j)}} \right) \frac{\partial \hat{y}}{\partial h_4} \frac{\partial L}{\partial \hat{y}} \\ &= 2(\hat{y} - y) w_5 f'(z_1) x \sum_{\ell=2}^4 \left(w_{\ell,1} \prod_{j \in \text{paths after } \ell} w_{j,j-1} f'(z_j) \right) \end{aligned} \quad (11)$$

In the CNN structure, with dense connectivity, the intermediate feature maps may have different numbers of channels. However, they must share the same spatial resolution. Consequently, dense connections are applied only within dense blocks, where the spatial dimensions are preserved, while changes in resolution are handled separately through transition layers.

2 ResNet

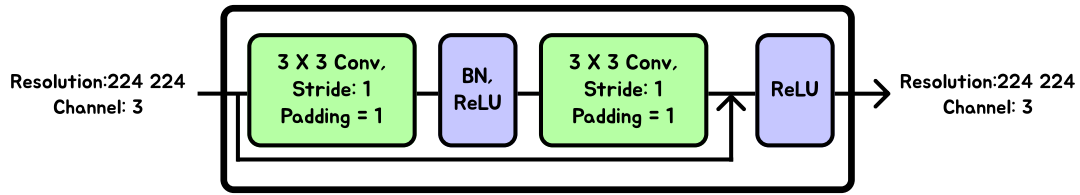


Figure 4: Resolution preserving residual block (18-layer or 34-layer ResNet)

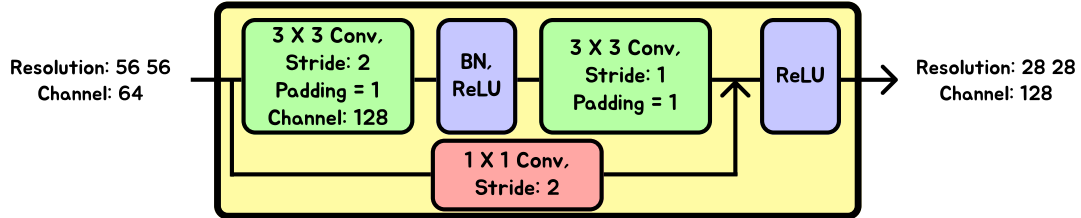


Figure 5: Resolution reducing residual block (18-layer or 34-layer ResNet)

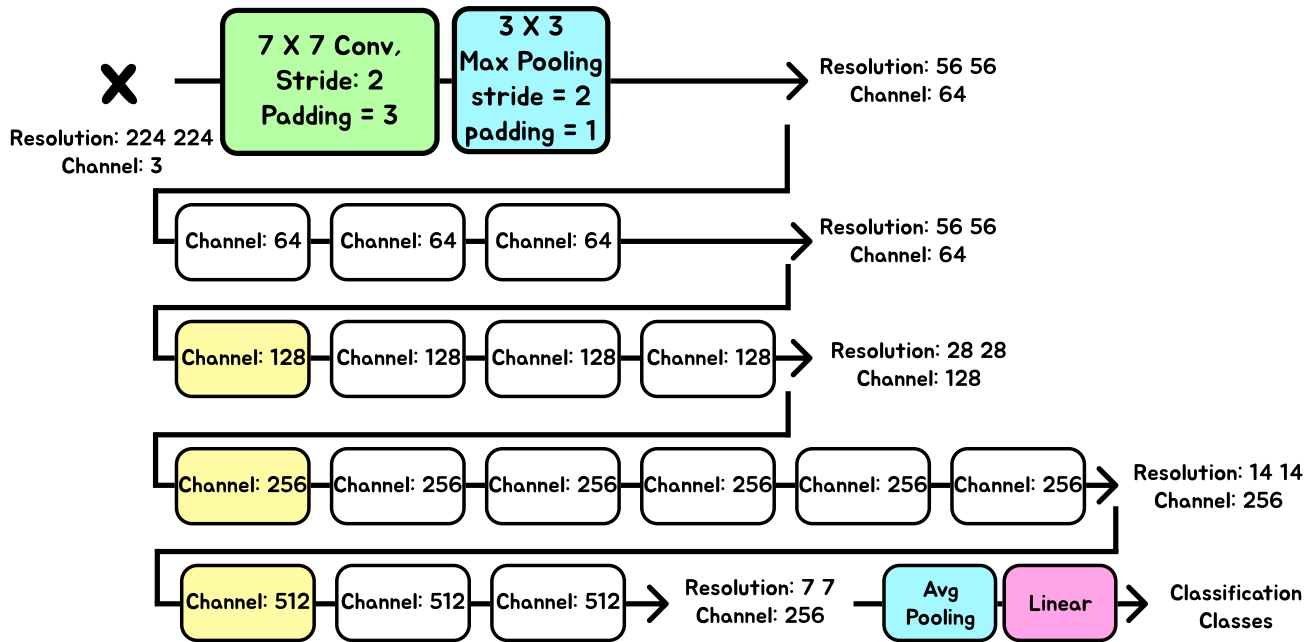
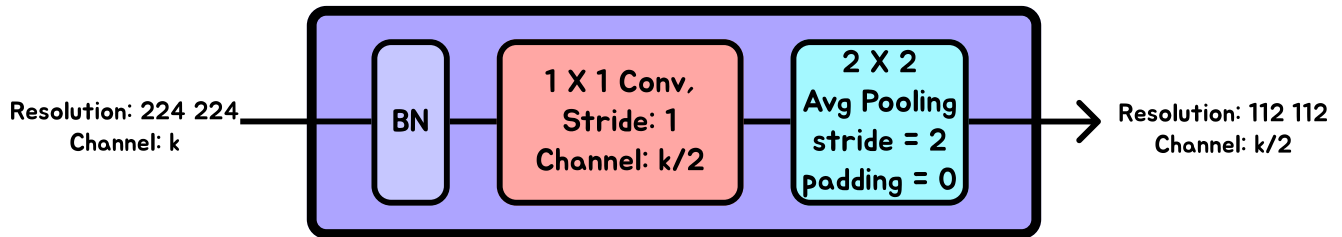
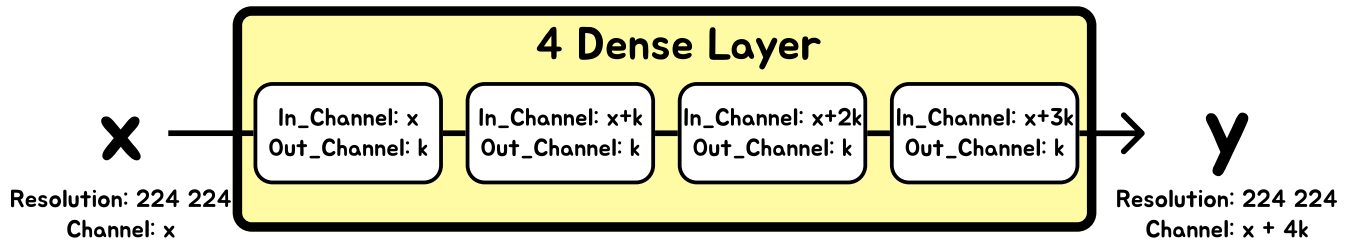
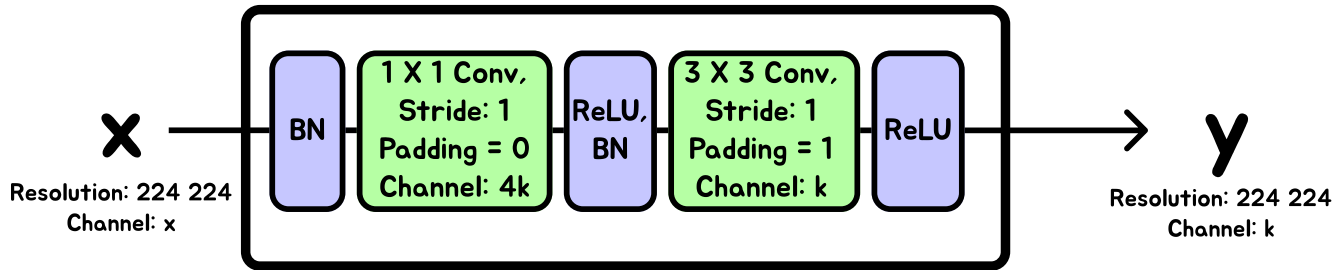


Figure 6: 34-layer ResNet

3 DenseNet



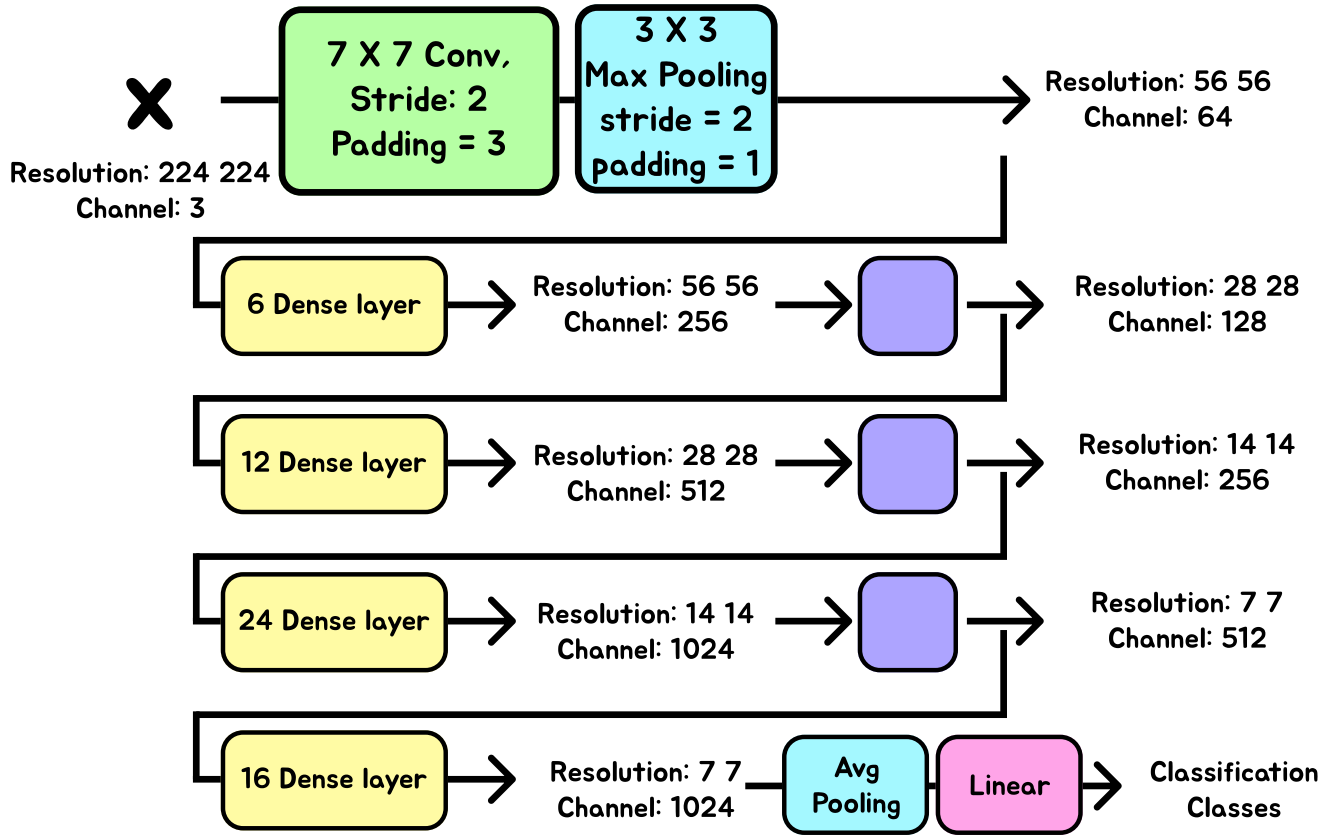


Figure 10: Structure of DenseNet-121.

References

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- [2] G. Huang, Z. Liu, L. Van Der Maaten, K. Q. Weinberger, Densely connected convolutional networks, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2017, pp. 4700–4708.